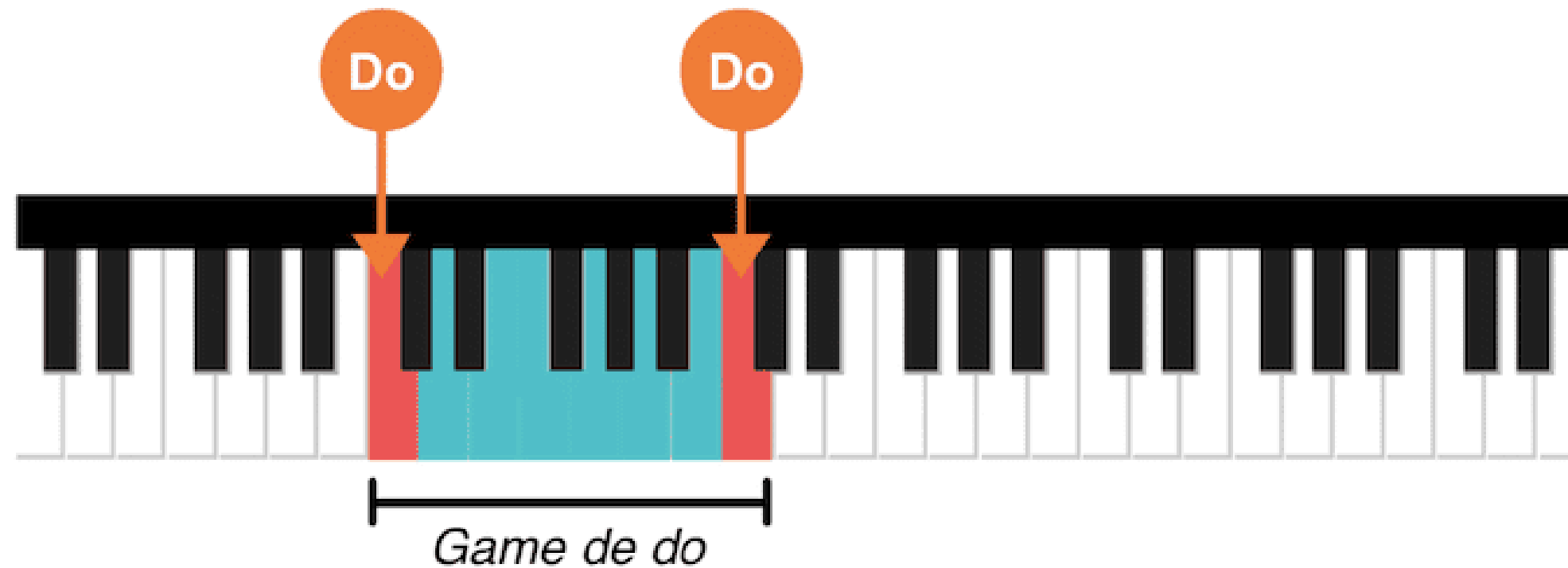


Developing a Digital Piano : One Octave Range



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TASK DISTRIBUTION

Project & Task Management : Léo Roupenel

Time Management & Deadlines : Julien Fauvel

Procurement & Ordering : Widad Nori

Testing & Quality Control : Wissem M'tir

Weekly Reports & Website : Elie Corbière

Final Report & Presentation : Claire Mestre Pincho

System Architecture

1. Acquisition

Convert a key press into an electrical signal readable by the microcontroller.

2. Processing

Analyze input signals in real-time and mathematically generate the corresponding audio signal (handled by the Nucleo board)

3. Power Supply

Provide a stable and safe voltage to all internal components.

4. Output

Amplify the electrical signal from processing and convert it into audible sound waves.

Initial Questions & Challenges

Acquisition

How to detect key presses across multiple keys ?

What type of buttons : Hall effect (magnetic) vs. mechanical switches ?

Processing

How to generate the audio signal ?

How to manage the note envelope (attack/decay) ?

How to handle polyphony ?

Power Supply

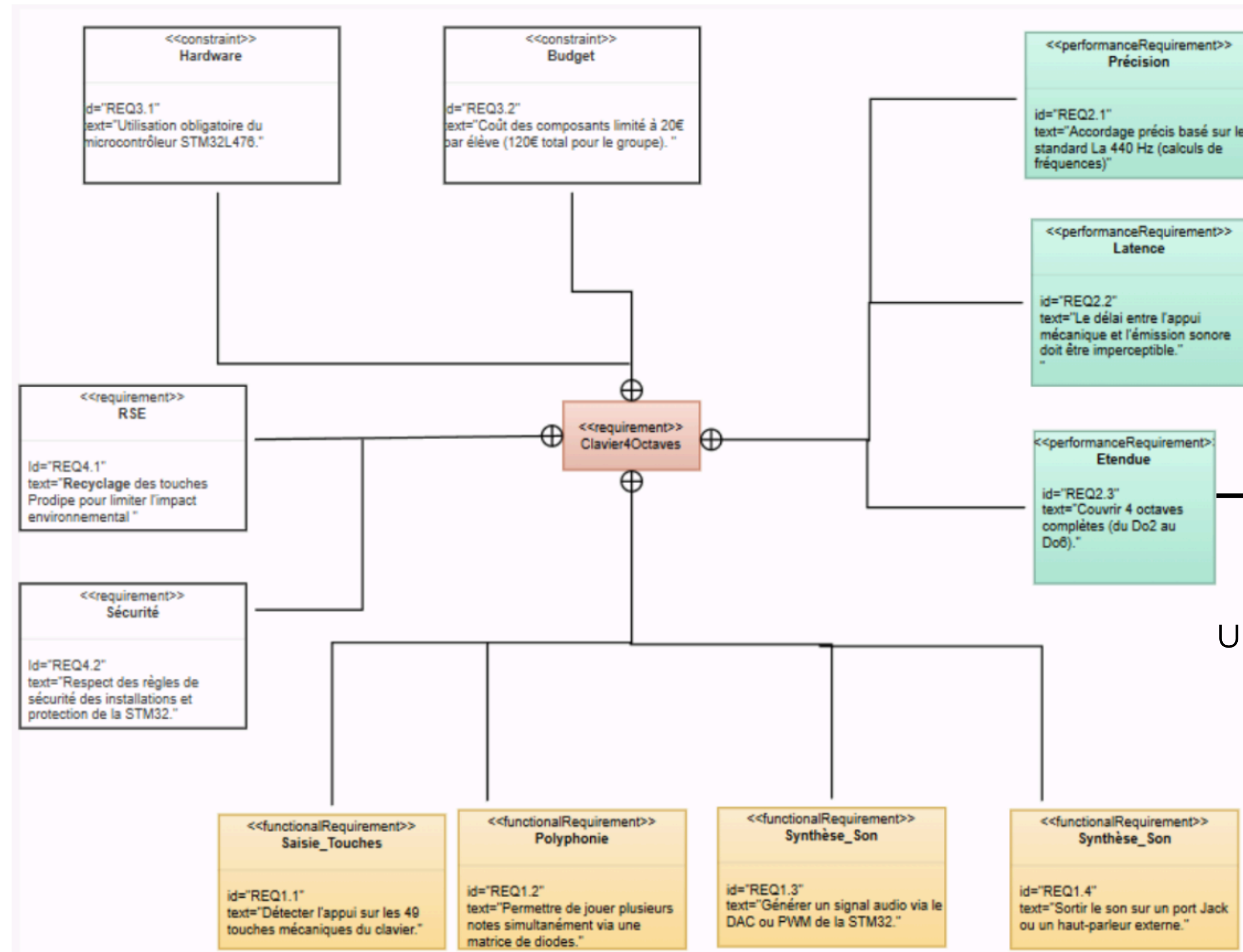
Battery (9V) vs. USB source ?

Output

Headphone jack, integrated speaker, or both ?

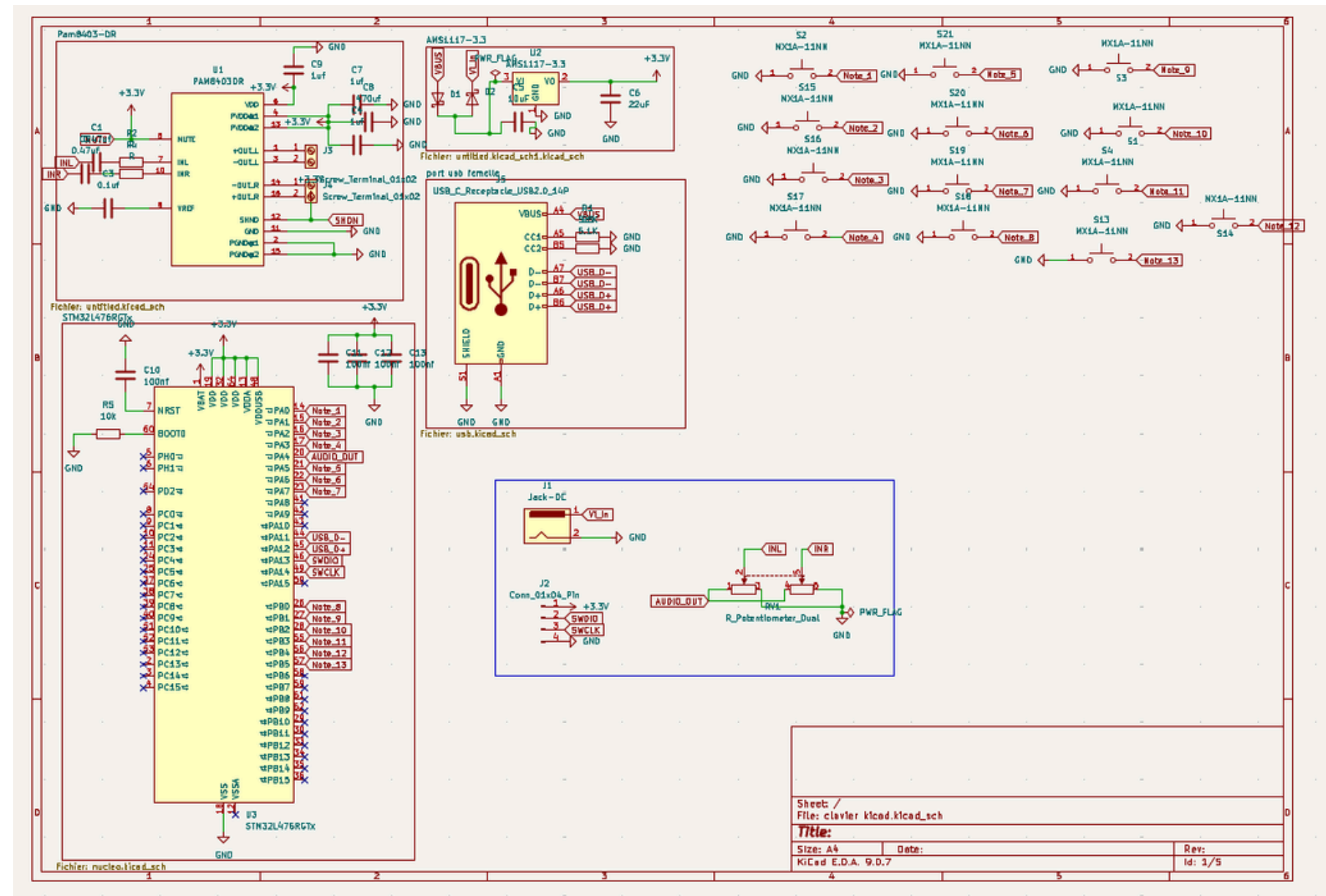
Bluetooth integration ?

Technical Specifications & SysML Requirements Diagram



Une octave (13 notes)

1. Acquisition



1. Acquisition : From questions to solutions

How to detect key presses across multiple keys ?

What type of buttons : Hall effect (magnetic) vs. mechanical switches (Cherry MX) ?

→ Selection of 13 Cherry MX Black Silent mechanical switches (one full octave, from C to C).

2. Processing : From questions to solutions

How to generate the audio signal ?

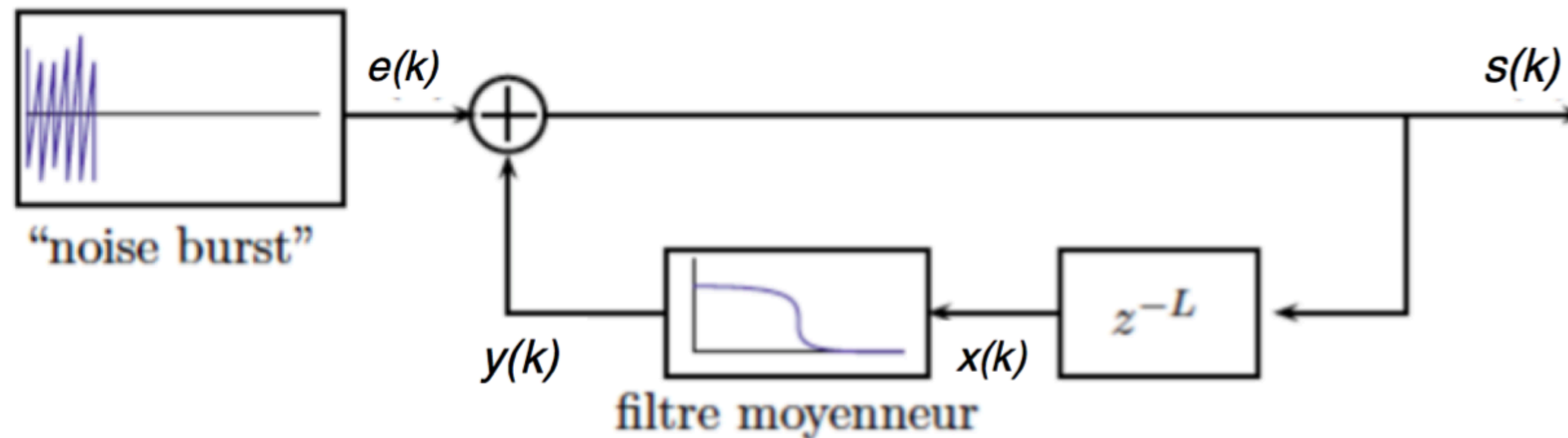
How to manage the note envelope (attack/decay) ?

How to handle polyphony ?

→ Sine wave generation via the integrated DAC.

→ Note envelope management (attack/decay) using an additive synthesis algorithm.

→ Prototype developed in Python then ported to C.



Karplus Strong algorithm in python

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.io.wavfile import write
```

```
sample_rate = 44100
```

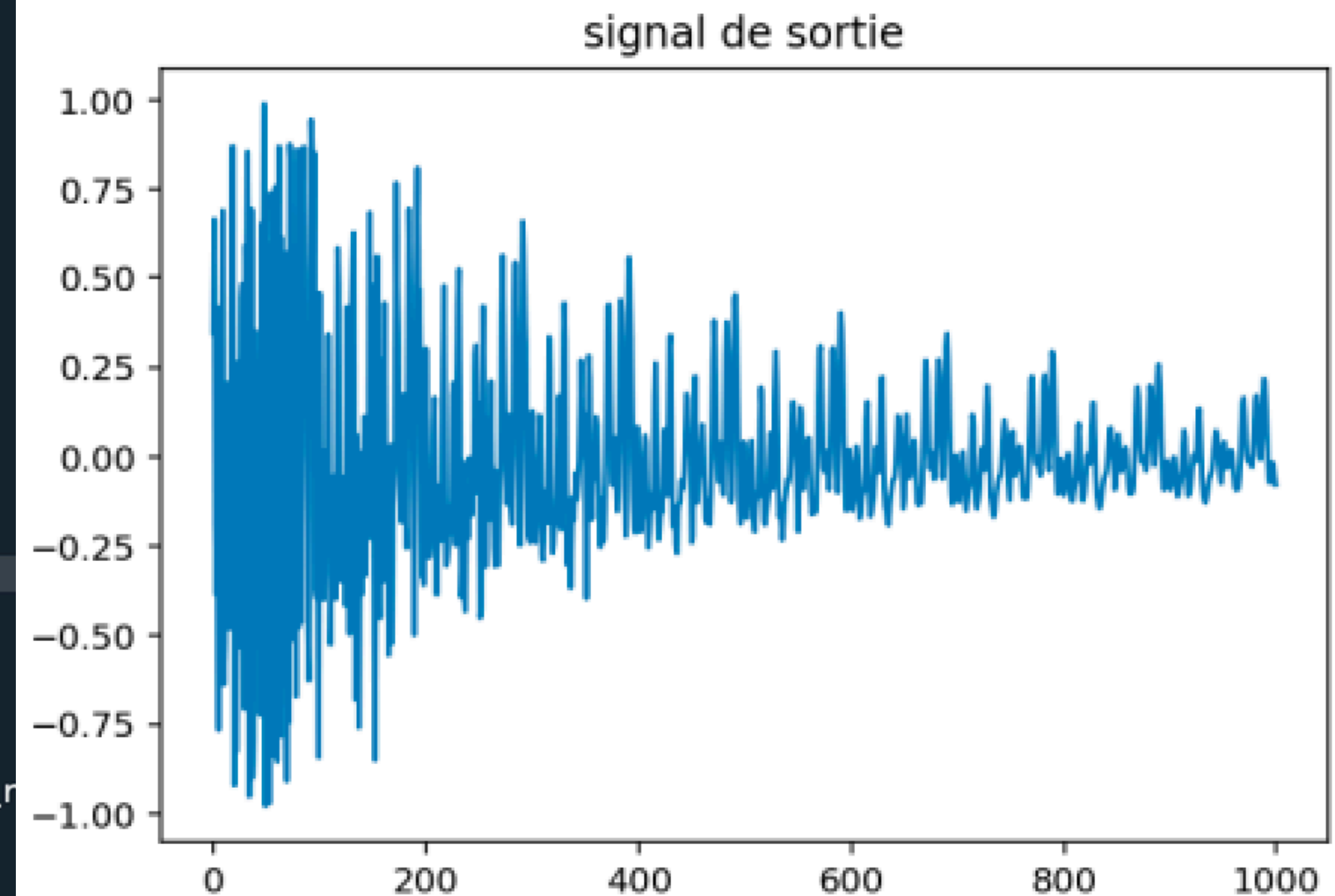
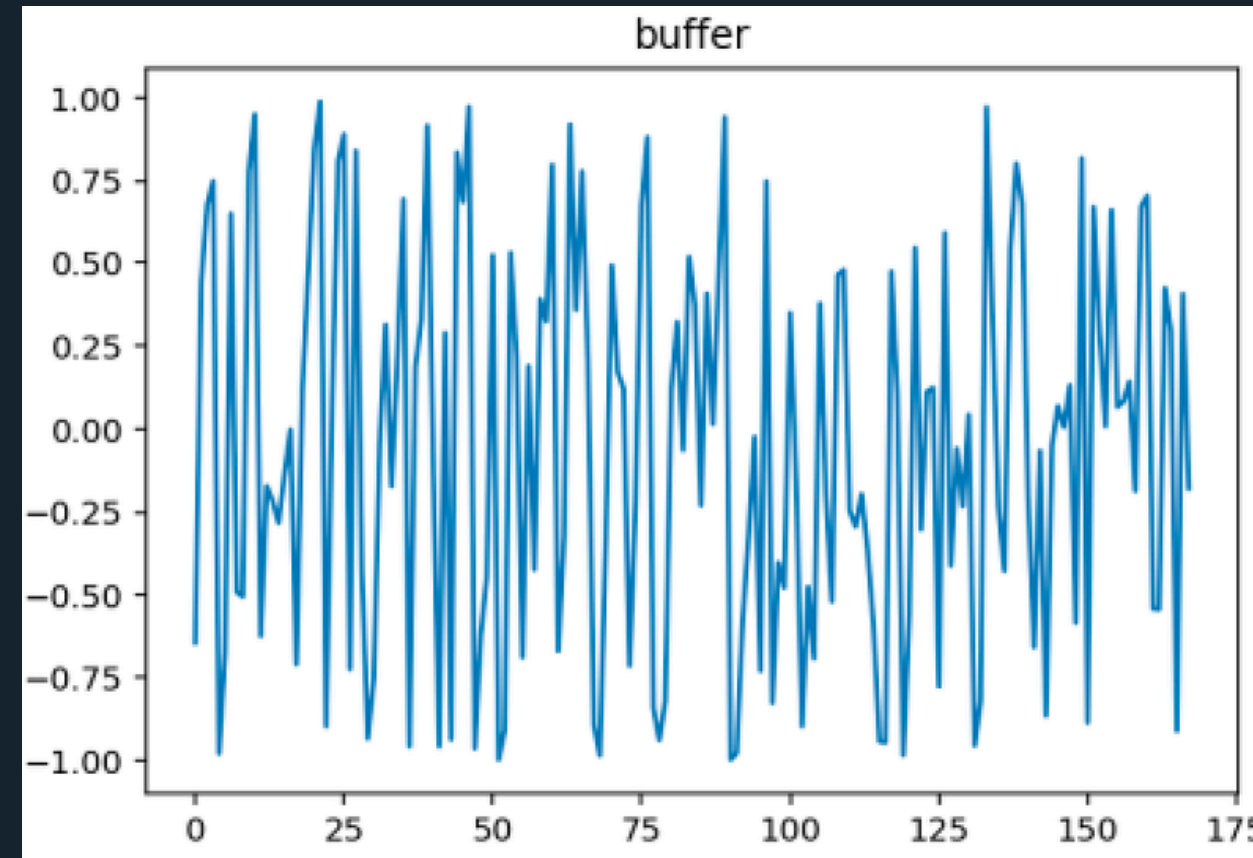
```
def bruit_blanc(nb_samples):
    return np.random.uniform(-1, 1, nb_samples)
```

```
def karplus_strong(frequency, duration, sample_rate=44100, damping=0.5):
    N = int(sample_rate / frequency)
    buffer = bruit_blanc(N)
    print('buffer', buffer)
    plt.plot(buffer)
    plt.title('buffer')
    plt.show()
    output = []
    for i in range(int(duration * sample_rate)):
        current = buffer[0]
        output.append(current)
        avg = damping * 0.5 * (buffer[0] + buffer[1])
        buffer = np.append(buffer[1:], avg)
    plt.plot(output)
    plt.title('signal de sortie')
    plt.show()
    return np.array(output, dtype=np.float32)
```

```
write("/Users/julienfauvel/Downloads/karplus_5.wav", sample_rate, karplus_strong(440, 2, sample_rate, 0.9))
```

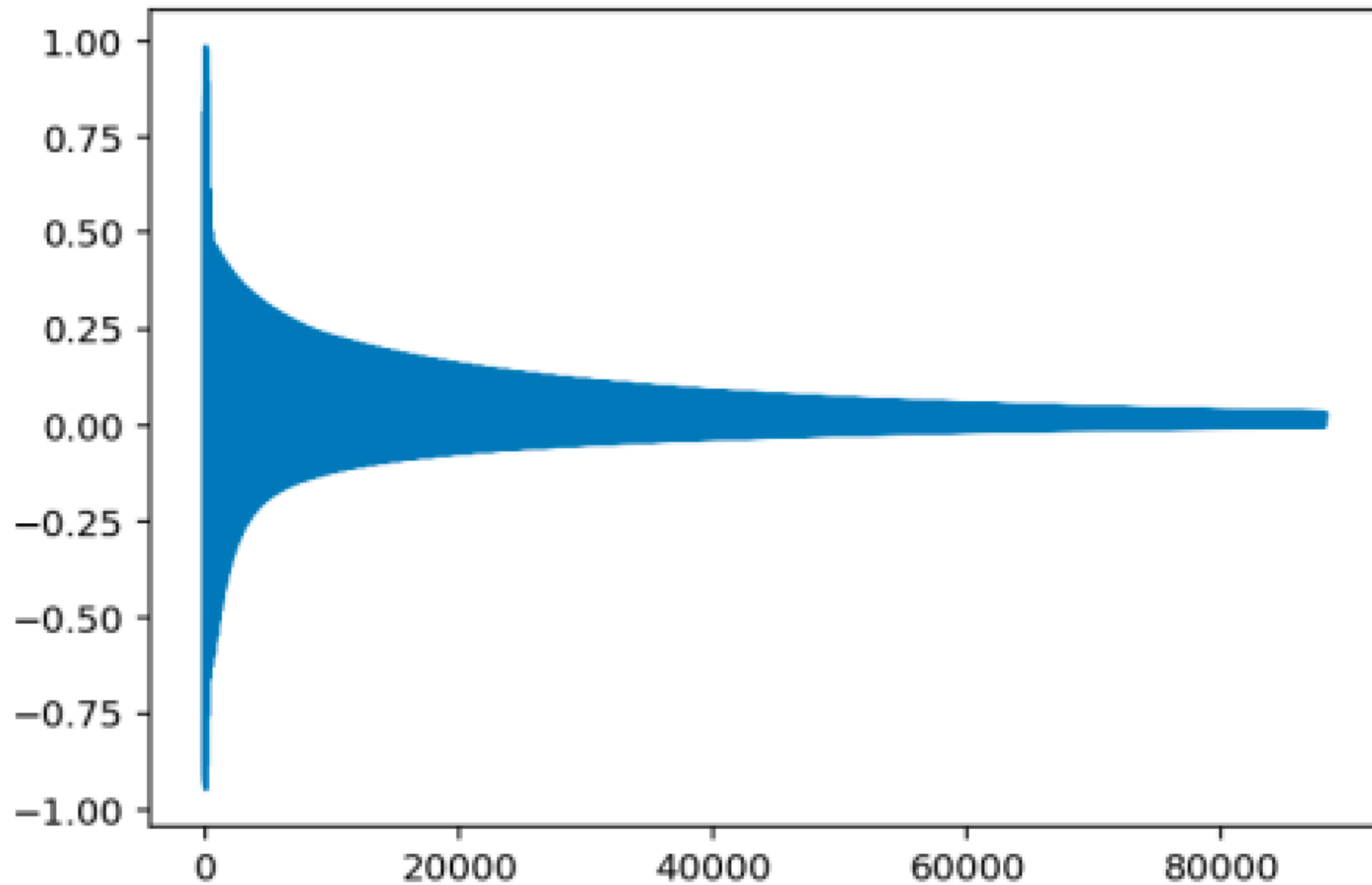
```
Do = 261.6
Re_b = 277.2
Re = 293.7
Mi_b = 311.1
Mi = 329.6
Fa = 349.2
Sol_b = 370
Sol = 392
La_b = 415.3
La = 440
Si_b = 466.2
Si = 493.9
```

```
def note_piano(note, duree):
    write("/Users/julienfauvel/Downloads/AAAAA1.wav", sample_rate, karplus_strong(note, duree+1.0, sample_rate, ((2*np.arctan(140*duree))/(np.pi))))
    print(2*np.arctan(140*duree)/(np.pi))
    return(karplus_strong(note, duree+1.0, sample_rate, ((2*np.arctan(140*duree))/(np.pi))))
```



Programming

signal de sortie



```
10 /**
11  * Initialise le buffer avec du bruit blanc [1].
12  * N = sample_rate / frequency
13  */
14 void KS_Init(KarplusStrong_t *ks, float frequency, float damping) {
15     // Calcul de N selon la source [1]
16     uint16_t N = (uint16_t)(SAMPLE_RATE / frequency);
17     if (N > MAX_BUFFER_SIZE) N = MAX_BUFFER_SIZE;
18
19     ks->size = N;
20     ks->index = 0;
21     ks->damping = damping;
22
23     // Remplissage avec du bruit entre -1.0 et 1.0 [1]
24     for (int i = 0; i < N; i++) {
25         ks->buffer[i] = ((float)rand() / (float)(RAND_MAX)) * 2.0f - 1.0f;
26     }
27
28 /**
29  * Calcule l'échantillon suivant (pas de l'algorithme)
30  * Correspond à : avg = damping * 0.5 * (buffer + buffer[1]) [1]
31  */
32 float KS_Step(KarplusStrong_t *ks) {
33     // Lecture de l'échantillon actuel
34     float current = ks->buffer[ks->index];
35
36     // Calcul de l'indice suivant pour le moyennage
37     uint16_t next_index = (ks->index + 1) % ks->size;
38
39     // Algorithme Karplus-Strong : Moyenne et atténuation [1]
40     float avg = ks->damping * 0.5f * (ks->buffer[ks->index] + ks->buffer[next_index]);
41
42     // Mise à jour du buffer (réinjection)
43     ks->buffer[ks->index] = avg;
44
45     // Avancement de l'index circulaire
46     ks->index = next_index;
47
48     return current;
49 }
50
```

3. Power Supply : From questions to solutions

Battery (9V) vs. USB source ?

→ USB-C source : More modern and reliable than 9V batteries.

→ AMS1117-3.3 Regulator : Ensures a stable 3.3V supply for sensitive components.

4. Output : From questions to solutions

Headphone jack, integrated speaker, or both ?

Bluetooth integration ?

→ Hybrid system with both an Integrated Speaker and a 3.5mm Jack.

→ Audio signal boosted by an Amplifier.

→ Volume control via a 10 k Ω Potentiometer.

→ Bluetooth excluded to avoid latency and simplify design.

List of components

1. Acquisition

- 13 boutons poussoirs

2. Processing

- Carte Nucléo-L476RG

3. Power Supply

- Connecteur d'entrée USB-C
- 2 résistances de configuration USB 5,1 k Ω
- Interrupteur
- Régulateur
- 1 condensateur de stabilisation de 10 μ F et 1 de 22 μ F

4. Output

- Amplificateur
- Composants passifs pour l'ampli :
 - 3 condensateurs de 1 μ F, 1 de 470 μ F, 2 de 0.47 μ F et 1 de 0.1 μ F
 - 2 résistances de 20 k Ω
- Haut-parleur
- Prise Jack
- Potentiomètre 10 k Ω

List of components

Fournisseur : Digikey					Envoyez un seul fichier par mail!			
Pos	I	P	Art	Désignation	Qté commandée	UA	Date Livraison	Prix unitaire
10	Z			Interrupteur R1973ABLK GILGF3	1	PC	18.03.2026	2.4
20	Z			Haut-parleur 1528-2752-ND	1	PC	18.03.2026	4.2
30	Z			Prise Jack 3,5 FC681375V	1	PC	18.03.2026	1.9

Fournisseur : Farnell				Envoyez un seul fichier par mail!				
Pos	I	P	Art	Désignation	Qté commandée	UA	Date Livraison	Prix unitaire
10	Z			Amplificateur 3373823	1	PC	23.03.2026	1.4
20	Z			Connecteur USB4105-GF-A 3588205	1	PC	23.03.2026	0.9

Fournisseur : Mouser					Envoyez un seul fichier par mail!			
Pos	I	P	Art	Désignation	Qté commandée	UA	Date Livraison	Prix unitaire
10	Z			Boutton-poussoir CherryMX1A-11NN	13	PC	23.03.2026	1.1
20	Z			Potentiomètre 10kOhm RK16312A0BKR	1	PC	23.03.2026	6.8

Project Timeline

SESSION 1

- **Project selection**
- **Initial project exploration (brainstorming)**

SESSION 2

- **Recovering a non-functional keyboard**
- **Analysis of internal keyboard design**
- **Harvesting keys for our project**
- **Establishing project specifications**

SESSION 3

- **Component selection**

SESSION 4

- **Finalizing the bill of materials**
- **Beginning of KiCad design phase**
- **Initial audio generation Programming (Python)**

SESSION 5

- **KiCad design finalization (in progress)**
- **Finalizing C code development**

SESSION 6

- **Mid-project progress review**
- **KiCad design completion**

SESSION 7

- Soldering
- Report drafting

SESSION 8

- Soldering
- Core competency (MC) self-assessment (Sessions & Others)
- Report drafting

SESSION 9

- Final report completion

SESSION 10

- Final core competency (MC) self-assessment (Oral & Written)
- Final report submission, archiving, and final defense with demos

Séance	MC1.1 - Comprendre le contexte et les spécifications du projet	2 pt	30% 6 points
	MC1.1 - Mobiliser des ressources variées pour trouver des informations et solutions pertinentes	2 pt	
	MC2.1 - Contribuer au projet de façon active	2 pt	
Oral	MC3.1 - Présenter le projet à l'oral avec un support complet et adapté	5 pt	25% 5 points
Écrit	MC3.1 - Rédiger un rapport écrit efficace et pertinent	5 pt	25% 5 points
Autres	MC3.1 - Tenir ses engagements : respect des tâches, échéances, ponctualité	2 pt	20% 4 points
	MC4.1 - Être sensibilisé aux contraintes réglementaires et de sécurité des installations et matériels utilisés dans le projet	1 pt	
	MC4.1 - Mener une réflexion sur une ou plusieurs thématiques RSE appliquées au projet : sociétal, éthique et/ou environnemental	1 pt	